

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow members to search the ISBN repository and place hold requests on currently loaned books to enter a FIFO waitlist.	High	Pass: User is assigned correct position in queue. Fail: User already holding max allowed books places an additional reservation.	Core reservation functionality enabling equitable, first-come-first-served access to high-demand titles.
FR-002	Functional	The system shall notify a Student Member by email/SMS within 5 minutes of a held book being checked in, and shall hold the copy at the circulation desk for 48 hours before releasing it to the next member in the queue.	High	Pass: Notification timestamp is within 5 minutes of check-in and item status changes to “Ready for Pickup”. Fail: Notification delayed beyond 5 minutes, or copy released before the 48-hour hold expires.	Ensures timely, fair access to holds and prevents an available copy from indefinitely blocking the queue.
FR-003	Functional	The system shall automatically calculate an overdue fine of ₹5 per day per book once the due date has passed, capping the total fine per title at ₹200.	Medium	Pass: Fine ledger accrues ₹5/day and stops incrementing at ₹200. Fail: Fine exceeds the cap or fails to accrue for an overdue title.	Automates fine tracking to reduce manual counter-staff errors and encourage timely returns.
FR-004	Functional	The system shall require Student Members and Head Librarians to authenticate with valid credentials (UC-00: Login / Authenticate) before accessing any personalized or restricted feature, such as placing holds, managing queues, or viewing fine details, and shall lock an account for 30 minutes after 5 consecutive failed login attempts within a 15-minute window.	High	Pass: Restricted features are inaccessible without a valid session, and the account locks after 5 failed attempts in 15 minutes. Fail: A restricted feature is reachable without authentication, or the account is not locked after the threshold is exceeded.	Protects member and staff data and ensures only authorized actors can reserve items, manage queues, or view sensitive fine information.
FR-005	Functional	The system shall block a Student Member from placing any new hold request while their total unpaid overdue fines exceed ₹500, and shall display the outstanding balance with a “Pay Now” prompt.	High	Pass: Hold request is rejected and correct outstanding balance is shown when fines exceed ₹500. Fail: Hold request succeeds despite fines exceeding the threshold.	Protects circulation of physical stock by discouraging chronic non-return and unpaid-fine behaviour.
NFR-001	Nonfunctional (Performance & Security)	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Guarantees a responsive search experience during peak periods such as semester start and exams.
NFR-002	Nonfunctional (Availability & Reliability)	The system shall maintain 99.5% monthly uptime for the catalog search and reservation services, excluding announced maintenance windows.	High	Pass: Monthly uptime logs show ≥99.5% availability eannounced maintenance. Fail: Unplanned downtime drops availability below 99.5%	Students and librarians depend on continuous, round-the-clock access to search, reserve, and manage holds.